

Measurement

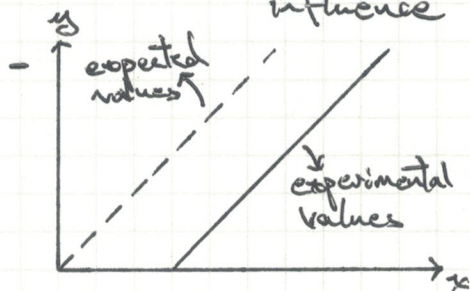
Uncertainty :

Accuracy : Degree of closeness to the true value of the quantity being measured.

Precision : Level of uncertainty in an instrument's measurement.

Systematic error

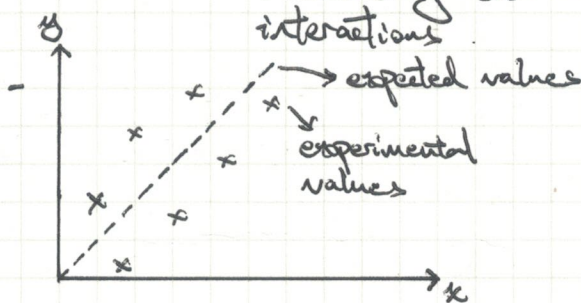
- Values lack accuracy
 - ↳ All data are off by a certain amount.
 - ↳ e.g. • Zero error
 - Constant environmental influence



- Solution : Identify error.

Random error

- Values lack precision
 - ↳ All values are off by random amount.
 - ↳ e.g. • Human errors
 - Inconsistencies in instrument
 - Fluctuating environmental interactions



- Solution : • Average a set of data.
- Plot a graph and draw best fit line.

Rules

1. Uncertainty is always 1 s.f.

2. Measured value has as many d.p. as its uncertainty.

+/- : Follow smallest d.p.

x/÷ : Follow smallest s.f.

Trigo : $Y = \sin X$, Y follow s.f. of

Log : $Y = \log_a X$, d.p. of y equal s.f. of x

Quantity

$$Q = aX \pm bY$$

$$Q = X^m \times Y^n$$

$$Q = X^m \div Y^n$$

Uncertainty

$$\Delta Q = a \Delta X + b \Delta Y$$

$$\frac{\Delta Q}{Q} = m \frac{\Delta X}{X} + n \frac{\Delta Y}{Y}$$

Units :

- * All valid equations in physics must be homogeneous.
- * Trigo functions take in angles as input and give a unit-less ratio.
- * Logarithmic functions have no units.

SI base quantities and units

Mass	-	kg
Length	-	m
Time	-	s
Electric current	-	A
Temperature	-	K
Amount	-	mol
Luminosity intensity	-	cd

Prefixes

pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

Common derived units in SI base units

Force	N	kg m s^{-2}
Energy	J	$\text{kg m}^2 \text{s}^{-2}$
Pressure	Pa	$\text{kg m}^{-1} \text{s}^{-2}$
Electric charge	C	A s
Electrical power	W	$\text{kg m}^2 \text{s}^{-3}$
Electrical potential difference	V	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$
Electrical resistance	Ω	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$

Scalar : A physical quantity that only has magnitude.

Vector : A physical quantity that has magnitude and direction.

Kinematics

Definitions :

Distance : Total length of path travelled.

Speed : Rate of change of distance.

Displacement : Total distance of a point from a reference point in specific direction.

Velocity : Rate of change of displacement. * $\Delta v = v + (-$

Acceleration : Rate of change of velocity.

Description of motion :

Body is at rest / moving in the positive / negative direction with a displacement / velocity / acceleration of (magnitude) that is increasing / decreasing / constant.

* When describing motion, state:

- ① Physical quantity
- ② Direction
- ③ Magnitude
- ④ Time evolution (Increasing / Decreasing / Constant)

Equations :

$$v = u + at$$

Derivation

$$- a = \frac{\Delta v}{\Delta t} = \frac{v - u}{t - 0}$$

$$s = \frac{1}{2} (u + v) t$$

$$- s = v_{\text{avr}} \times t = \left(\frac{u + v}{2} \right) t$$

$$s = ut + \frac{1}{2} at^2$$

$$- \text{Integration: } \int v dt = \int u + at dt$$

$$v^2 = u^2 + 2as$$

$$- \text{Sub } t = \frac{v - u}{a} \text{ into } s = ut + \frac{1}{2} at^2$$

* Motion in perpendicular directions are completely independent.

Free fall : $ma = mg \Rightarrow a = g$
 $F_{\uparrow} = F_{\downarrow}$

Dynamics

Newton's Laws of motion :

Newton's First Law : An object at rest will remain at rest and an object in motion will remain in motion at constant velocity in the absence of an external resultant force.

Inertia : Reluctance of a body to change its state of rest or uniform motion in a straight line.

Mass : Measure of a body's resistance to a change in velocity.

Weight : Force acting on a mass due to a gravitational field.

Newton's Third Law : If body A exerts a force on body B, body B exerts a force of the same type that is same in magnitude and opposite in direction on body A.

Characteristics of action-reaction pairs

1. Same type of force
2. Same magnitude but opposite in direction
3. Act on different bodies

* A closed system contains all action-reaction pairs

* Action-reaction pairs occur at the same time

Newton's Second Law : Rate of change of momentum of a body is directly proportional to the resultant force and occurs in the same direction as the resultant force.

$$F_{\text{net}} \propto \frac{dp}{dt}$$

$$\Rightarrow F_{\text{net}} = \frac{dp}{dt} = \frac{dm}{dt} v + m \frac{dv}{dt}$$

Linear momentum : Product of the mass and velocity.

* Linear momentum occurs in the same direction as the velocity.

* Objects in a system : $\Sigma F = ma$

Momentum :

Impulse : Product of an average force F_{avr} acting on an object and the time Δt for which the force acts.

$$\text{Impulse} = F_{avr} \Delta t = \int_{t_i}^{t_f} F dt = \Delta p$$

Applications :

- Constant force, increased time of contact $\rightarrow$ larger final momentum
e.g. When kicking a ball, a follow-through increases time of contact hence increasing final momentum.
- Constant impulse, increased time of contact $\rightarrow$ smaller force
e.g. Air bags in cars:
During a crash, hitting an air bag increases the time of contact hence reducing the force of impact.
- Landing a jump:
Bending the knees reduces impact force by increasing the time taken to come to a complete stop.
- Accelerating car/jet:
For the same change in momentum, taking a longer time to accelerate exerts a smaller force on the driver/pilot, reducing strain and injury.

Principle of Conservation of Linear Momentum : Total momentum of a system remains constant provided no external resultant force acts on the system.

Initial momentum = Final momentum

Elastic collision : Collision in which total kinetic energy is conserved.
* In an elastic collision, relative speed of approach equal relative speed of separation.
 $u_1 - u_2 = v_2 - v_1$

Inelastic collision : Collision in which total kinetic energy is not conserved.

* Momentum is still conserved in inelastic collisions.

* For cases where internal energy is released, $\frac{KE_1}{KE_2} = \frac{|v_1|}{|v_2|} = \frac{m_2}{m_1}$

Perfectly inelastic collision : Colliding bodies stick together and move on with

Forces

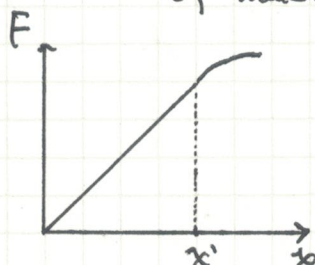
Tension :

Hooke's Law : Restoring force is directly proportional to the extension / compression in a material if its limit of proportionality is not exceeded.

$$F = kx$$

$$x = L - L_0$$

* k : measure of the stiffness of material



* Spring constant, K :

Series : $\frac{1}{K_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$

Parallel : $K_{eq} = k_1 + k_2 + \dots$

* Energy :

Series : $\frac{E_1}{E_2} = \frac{\frac{1}{2} k_1 x_1^2}{\frac{1}{2} k_2 x_2^2} = \frac{x_1}{x_2} = \frac{k_2}{k_1}$

Parallel : $\frac{E_1}{E_2} = \frac{\frac{1}{2} k_1 x^2}{\frac{1}{2} k_2 x^2} = \frac{k_1}{k_2}$

Upthrust : Net force that arises due to fluid pressure.

* Equal in magnitude and opposite in direction to the weight of the fluid displaced by a submerged or floating object.

Pressure : Force per unit area.

$$P = \frac{F}{A} \quad * \text{Fluid pressure} = \frac{F}{A} = \frac{mg}{A} = \frac{(\rho h A)g}{A} = \rho g h$$

Density : Mass per unit volume.

$$F_b = \rho g h A = \rho g V$$

$$P_{\text{top surface}} = \rho g h_1$$

$$P_{\text{bottom surface}} = \rho g h_2$$

$$F = \rho g (h_2 - h_1) A = \rho V g$$

* ρ and V are of the fluid displaced.

F_b = weight of fluid displaced

* Principle of floatation : F_b = Weight of object when system is in equilibrium

Centre of gravity : Point at which the whole weight of a body may be thought to act.

* A beaker of fluid has weight Z . A solid of weight X displaces weight Y of fluid when fully immersed. When placed on a weighing scale,



* When an ice cube is floating on water, to support its mass, upthrust causes volume of water measured to be higher.

Moment :

Moment of a force : Product of the force and the perpendicular distance of the line of action of the force from the pivot.

$$\text{Moment} = F r \sin \theta$$

Torque : Product of one of the forces and the perpendicular distance between the lines of action of the forces.

$$\tau = Fd = 2Fr$$

* A couple is a pair of forces equal in magnitude and opposite in direction.
 $\Rightarrow$ No linear motion but a net turning effect is produced.

Principle of Moments : If a body is in (rotational) equilibrium, the sum of all clockwise moments about any axis must equal the sum of all anticlockwise moments about the same axis.

Equilibrium :

Conditions :

1. Resultant force acting on body = 0 (Translational Equilibrium)

2. Resultant moment about any point = 0 (Rotational Equilibrium)

* If lines of action of the forces intersect at one point, the system is in rotational equilibrium.

Work, Energy and Power

Work : Product of a force and the displacement in the direction of the force.

One joule of work : Work done by the force of 1 newton when its point of application moves 1 m in the direction of the force.

$$W = F s \cos \theta$$

$$= \int F ds$$

= Area under F-s graph

* Work done by expanding gas = $p(V_f - V_i)$

Work done by spring = $\frac{1}{2} k x^2$ = $\int_{V_i}^{V_f} p dV$

Energy : Ability to do work.

Principle of Conservation of Energy : Energy can neither be created nor destroyed in any processes.

$$F = - \frac{dU}{dx}$$

Kinetic energy :

$$W = F s = m a s$$

$$= m \left(\frac{1}{2} v^2 - u^2 \right)$$

$$KE = \frac{1}{2} m v^2$$

Work-Energy Theorem : Net work done by external forces acting on an object is equal to the change in kinetic energy of the particle.

$$W = \Delta KE$$

Gravitational potential energy :

$$W = F s = m g (h_f - h_i)$$

$$GPE = m g h$$

* Work done by conservative forces (gravity, tension, electric) are independent of path.

Power : Rate of doing work.

Average power : Total work done divided by total time taken.

$$\langle P \rangle = \frac{W}{t} = F \langle v \rangle$$

$$\text{Instantaneous power} = F v$$

$$\text{Efficiency } \eta = \frac{\text{Effective energy / power}}{\text{Total energy / power}}$$

Temperature and Ideal gas

Temperature :

Zeroth Law of Thermodynamics : If bodies A and B are separately in thermal equilibrium with body C, then bodies A and B are in thermal equilibrium with each other.

Absolute zero = $0\text{ K} = -273.15^\circ\text{C}$, where $p = 0$, $V = 0$

Ideal gas law : $pV = nRT$

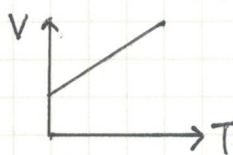
* $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$

Avogadro's constant N_A : The number of atoms in 0.012 kg of carbon-12.

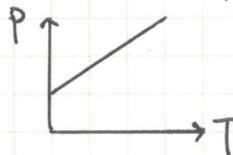
Boyle's Law : $pV = k$
when T is constant



Charles' Law : $\frac{V}{T} = k$
when p is constant



Gay-Lussac's Law : $\frac{p}{T} = k$
when V is constant



Assumptions

1. Gas particles have negligible intermolecular forces of attraction.
2. Gas particles occupy negligible volume compared to the container.

Pressure of ideal gas :

Let there be a gas with N molecules of mass m moving randomly in a cubic container of side L .

Consider a particle of speed v_x .

$$\Delta p = 2mv_x$$

$$\Delta t = \frac{2L}{v_x}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{mv_x^2}{L}$$

$$p \text{ due to one molecule} = \frac{F}{A} = \frac{mv_x^2}{V}$$

Let $\langle v_x^2 \rangle$ be the r.m.s. velocity in the x -direction.

$$p = \frac{Nm\langle v_x^2 \rangle}{V} = \frac{Nm\langle v^2 \rangle}{3V}$$

Kinetic energy of ideal gas :

One particle : $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$

Entire gas : $\frac{1}{2} N m \langle c^2 \rangle = \frac{3}{2} N kT = \frac{3}{2} nRT = \frac{3}{2} pV$

* Mean KE $\propto T$

Total KE $\propto NT$

* Monoatomic gas : $U = KE = \frac{3}{2} NkT$

Non-monoatomic gas : $U \propto NT$

$$U \neq \frac{3}{2} NkT$$

First Law of Thermodynamics

Heat capacity, C : Amount of thermal energy required to raise a body's temperature by 1 K.

Specific heat capacity, c : Amount of thermal energy required to raise 1 kg of a body's temperature by 1 K.

$$Q = mc(\theta_f - \theta_i)$$

Specific latent heat, L : Amount of thermal energy required to convert 1 kg of a substance to another phase without a change in temperature.

$$Q = mL$$

* Q_{vapor} is high as work needs to be done against atm. pressure.

* Thermal energy supplied by Heater = Thermal energy gained by material assuming 100% efficient

Internal energy : Sum of a random distribution of kinetic and potential energies associated with the molecules of a system.

* Potential energy of an ideal gas = 0 $\Rightarrow$ Internal energy $U = KE \propto T$

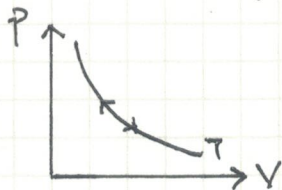
First Law of Thermodynamics : The increase in internal energy of a closed system is the sum of the heat supplied to the system and the work done on the system.

$$\Delta U = Q + W$$

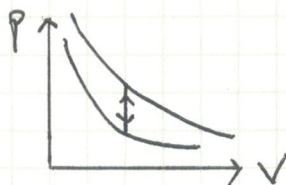
Thermodynamic processes

Isothermal process : Constant temperature process

* Since U of ideal gas $\propto T$, $\Delta U = 0 \Rightarrow Q = W$



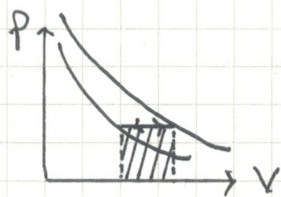
Isochoric process : Constant volume process



$$W = 0$$

$$\Delta U = Q$$

Isobaric process : Constant pressure process



$$W = p(V_f - V_i)$$

Adiabatic process : Process with no heat transfer in or out of a system.



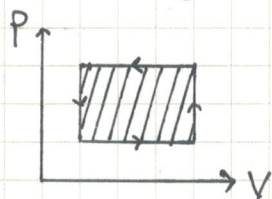
$$Q = 0$$

$$\Delta U = W$$

* Achieved when:

- System is insulated
- Change in pressure and volume occur faster than the exchange of heat

* For cyclic processes, $\Delta U = 0 \Rightarrow Q = W$



$$\Delta U = 0$$

Net work done on/by gas = Area enclosed by loop

Circular motion

* For any circular motion to exist, there must be a resultant centripetal acceleration.

Non-uniform circular motion

* Tangential speed is not constant.

* Use conservation of energy to solve questions.

Uniform circular motion

* Tangential speed is constant, no tangential acceleration.

Angular displacement : Angle through which an object turns with respect to the centre of the circular path.

$$\theta = \frac{s}{r}$$



Radian : Angle subtended at the centre of the circle by an arc length equals to the radius of the circle.

Angular velocity : Rate of change of angular displacement.

$$\omega = \frac{d\theta}{dt}$$

* ω is constant for uniform circular motion.

$$= \frac{2\pi}{T} = 2\pi f$$

$$v = r\omega$$

Centripetal acceleration $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$

* Since centripetal acceleration is always perpendicular to velocity, it only affect the direction and not the magnitude.

Centripetal force $F_c = m\frac{v^2}{r} = mr\omega^2 = mv\omega$

* Centripetal force does no work.

Application :

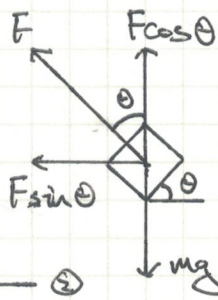
(1) Banked curve

$$F \cos \theta = mg \quad \text{--- (1)}$$

$$F \sin \theta = m r \omega^2 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} : \tan \theta = \frac{r\omega^2}{g} = \frac{v^2}{gr}$$

$$\therefore \theta = \tan^{-1}\left(\frac{v^2}{gr}\right)$$

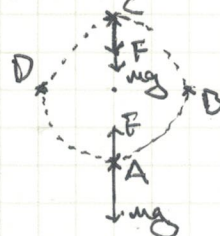


(2) Vertical uniform circular motion

$F_c \Rightarrow$ constant for ω to be constant.

$$F_c \text{ at C} = F + mg$$

$$F_c \text{ at A} = F - mg$$



Maximum F occurs at A.

Minimum F occurs at C.

Gravitation

Newton's Law of Gravitation : The force of attraction between any two point masses is directly proportional to the product of their masses and inversely proportional to the square of their separation.

$$F = -G \frac{m_1 m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

* Negative sign : Force is attractive in nature.

* Forces are action - reaction pairs.

* Spherical objects can be treated as point masses if :

- Separation $\gg$ Radius OR
- Homogeneous spheres OR
- Mass is distributed symmetrically in layers

* Apparent weightlessness : $N = 0$

True weightlessness : $F_{\text{net}} = 0$

Kepler's Third Law :

$$F_G = F_{\text{centripetal}}$$

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

$$= m r \omega^2$$

$$T^2 = \frac{4\pi^2}{GM} r^3$$

Geostationary orbit :

- West to East
- Directly above the Equator
- Period = 24 hours

⊕ : - Receiving and transmitting antennas can be fixed in place.

- Geostationary satellites are high in altitude allowing them to cover a large area.

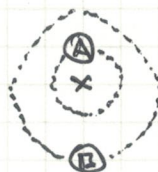
⊙ : - Signals take a long time to travel.

- Hard to send up.

- Only one orbit exists.

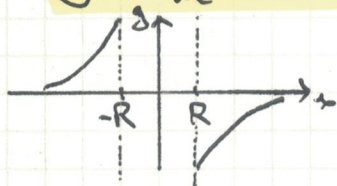
Binary stars :

$$F_G = F_{CA} = F_{CB}$$



Gravitational field strength : Gravitational force per unit mass acting on a small mass placed at the point.

$$g = \frac{F}{m} = -\frac{GM}{r^2} = -\frac{d\phi}{dr}$$



* Negative sign : Direction of field is towards while increasing r is taken as positive

* Gravitational field : Region of space where a mass experiences a gravitational force.

Gravitational potential energy : Work done by an external agent in bringing the mass from infinity to the point without any change in kinetic energy.

$$U_G = -\frac{GMm}{r}$$

* Gravitational potential energy at infinity = 0

$$F_G = -\frac{dU}{dr}$$

* Since F_G is attractive in nature, work done by force is opposite in direction from the mass.

* The further away from the mass the higher the potential energy.

Gravitational potential : Work done per unit mass by an external force in bringing the mass from infinity to the point without a change in kinetic energy.

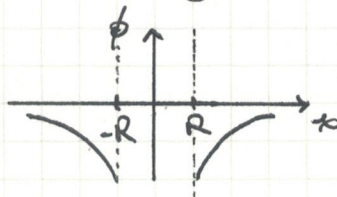
$$\phi = \frac{U_G}{m} = -\frac{GM}{r}$$

* Scalar

* Gravitational potential at infinity = 0

$$\Delta\phi = \phi_{\text{final}} - \phi_{\text{initial}}$$

$$* g = -\frac{d\phi}{dr}$$



Energy of masses in orbit :

$$\text{GPE} = -\frac{GMm}{r}$$

$$\text{KE} = \frac{GMm}{2r}$$

$$\text{Total energy} = -\frac{GMm}{2r}$$

$$* \text{KE} = \frac{1}{2}mv^2$$

$$F_c = F_g$$

$$m\frac{v^2}{r} = G\frac{Mm}{r^2}$$

$$v^2 = G\frac{M}{r}$$

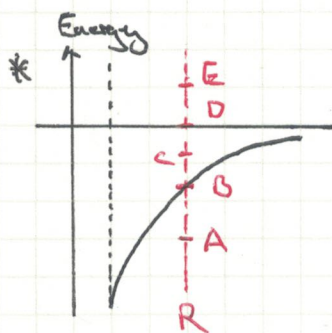
$$\Rightarrow \text{KE} = \frac{GMm}{2r}$$

Escape velocity :

Minimum KE = Gain in GPE

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

$$v = \sqrt{\frac{2GM}{r}}$$



A : Impossible

B : Rocket is stationary

C : Rocket has KE half of GPE. It could be in orbit.

D : Rocket will move to infinity and stationary at infinity.

E : Rocket will reach infinity with non-zero KE.

* Apparent weight = True weight - Centripetal force

* Distance between objects in space is Center-to-center separation!

Oscillations

Simple harmonic motion : Motion of a body whose acceleration is directly proportional to its displacement from a fixed point and is always directed towards that fixed point.

* Negative sign indicates that direction of acceleration is always opposite of x .

$$* ma = -kx$$

$$\Rightarrow a = -\frac{k}{m}x$$

Displacement x : Distance of oscillating object from the equilibrium position at any instant.

Amplitude x_0 : Maximum displacement from the equilibrium position.

Period T : Time taken to complete one cycle of oscillation.

Frequency f : Number of cycles per unit time made by the oscillating object.

$$\text{Angular frequency } \omega = 2\pi f = \frac{2\pi}{T}$$

Phase ϕ : Angle which gives a measure of the fraction of a cycle that has been completed by an oscillating particle or wave.

Phase difference $\Delta\phi$: Measure of how much one wave is out of step with another.

* In phase :

$$\Delta\phi = 0, 2\pi, 4\pi \dots$$

Antiphase :

$$\Delta\phi = \pi$$

$$\frac{\Delta x}{\lambda} = \frac{\Delta\phi}{2\pi} = \frac{\Delta t}{T}$$

Kinematics : $x = x_0 \sin(\omega t)$

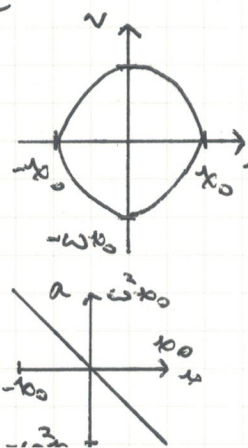
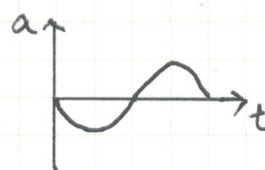
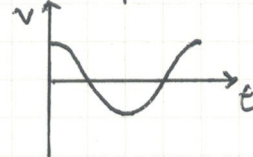
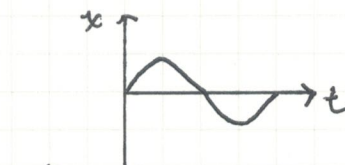
* Read question for equation!!!

$$v = \omega x_0 \cos(\omega t)$$

$$= \pm \omega \sqrt{x_0^2 - x^2}$$

$$a = -\omega^2 x_0 \sin(\omega t)$$

$$= -\omega^2 x$$

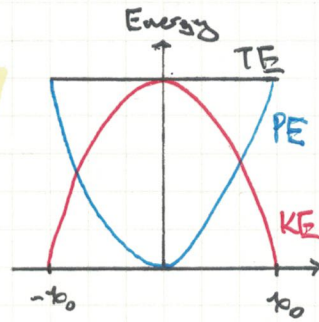


Energy :

$$\text{Kinetic energy} = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$$

$$\text{Potential energy} = \frac{1}{2} k x^2 = \frac{1}{2} m \omega^2 x^2$$

$$\text{Total energy} = \frac{1}{2} m \omega^2 x_0^2$$



Types of oscillations

Free oscillation : A freely oscillating system oscillating at its own natural frequency without external influences other than the initial impulse when displaced from its equilibrium position with no dissipation of energy.

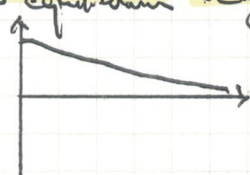
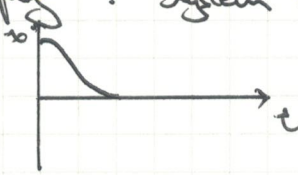
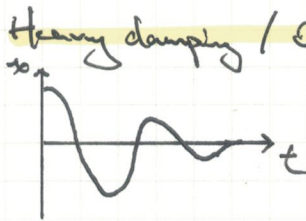
Natural frequency : Frequency at which an object will vibrate when it is allowed to do so freely.

Damped oscillations : Oscillations in which the amplitude diminishes with time as a result of dissipative forces that reduce the total energy of the oscillations.

* **Light damping / Underdamping** : Amplitude decreases exponentially. Period becomes slightly longer.

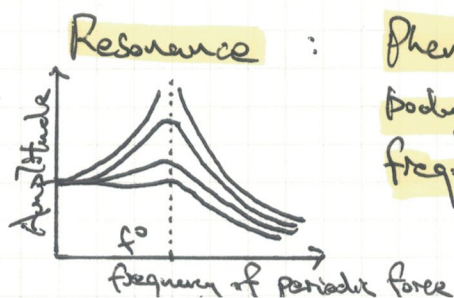
Critical damping : System returns to equilibrium in the shortest time without oscillating.

Heavy damping / Overdamping : System returns to equilibrium very slowly.



Forced oscillations : A system forced to oscillate at a frequency other than the natural frequency by a periodic external force.

Forced frequency : Frequency at which an object is made to vibrate by imposing a periodic force on it.



Resonance : Phenomenon that occurs when the frequency at which body is made to vibrate is equal to its resonant frequency of vibration.

* **Application** : Good - Music harmonics, Radio amplifier

Wave motion

Waves

Progressive waves : Waves in which energy is carried from one point to another by means of vibrations or oscillations within the wave.

Transverse waves : Waves in which the plane of oscillation of particles is perpendicular to the direction of transfer of energy of the wave.

Longitudinal waves : Waves in which displacement of particles are along the direction of transfer of energy of the wave.

Displacement : Distance moved by a particle in a wave in a specific direction from its equilibrium position.

Amplitude A : Magnitude of the maximum displacement of an oscillating particle from its equilibrium position.

Period T : Time elapsed between successive occasions when a particle is moving through the same point in the same direction.

Frequency f : Number of crests that pass a given point per unit time interval.

Wavelength λ : Shortest distance between two points on a progressive wave that are in phase, or the distance travelled by the wave energy during one complete oscillation of the source.

Wavefront : Locus of points having the same phase.

Speed of waves : Waveform advances a distance of λ in one period

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

$$v = \frac{\lambda}{T} = f\lambda$$

Phase difference :
$$\frac{\Delta x}{\lambda} = \frac{\Delta t}{T} = \frac{\Delta \phi}{2\pi}$$

Intensity : Wave energy incident per unit time per unit area normal to the wave.

$$I = \frac{\text{Power}}{\text{Area}}$$

$$I \propto E$$

$$I \propto (\text{Amplitude})^2$$

* Assuming no loss in energy!

Spherical waves : $I \propto \frac{1}{r^2}$

* Surface area of sphere
 $= 4\pi r^2$

$$I = \frac{P_0}{4\pi r^2}$$

Circular waves : $I \propto \frac{1}{r}$

Plane waves : $I = k$

Polarisation : When oscillations in a wave are confined to one direction only.

* Only transverse waves can be polarised.

Malus's Law : Intensity I of a beam of plane-polarised light after passing through a rotatable polariser is directly proportional to the square of the cosine of the angle θ through which the polariser is rotated from the position that gives maximum intensity.

$$I = I_{\text{max}} \cos^2 \theta$$

$$\Rightarrow I \propto \cos^2 \theta$$

* Plane polarised EM waves have their electric field oscillate in one direction.

Superposition

Principle of Superposition : When two or more waves of the same type meet at a point in space, the resultant displacement of the waves at any point is the vector sum of the displacement due to each wave acting independently at that point.

Interference : Superposing of two or more waves to produce regions of maxima and minima in space.

* **Coherence** : Sources with constant phase difference with respect to time.

* **Conditions for observable interference pattern**

1. Waves must overlap to produce regions of maxima and minima.

2. Sources must be coherent.

3. Waves must have the same / approximately the same amplitude.

4. Waves must be unpolarised / have the same plane of polarisation.

* **Path difference** $\Delta x = |S_1P - S_2P|$

S_1 and S_2 : Maxima Minima
are in phase : $\Delta x = n\lambda$ $\Delta x = (n + \frac{1}{2})\lambda$

S_1 and S_2 : a rad out of phase : $\Delta x = (n + \frac{1}{2})\lambda$ $\Delta x = n\lambda$

Stationary waves : Waves whose waveform does not advance and there is no translocation of energy.

* Stationary waves are formed when two progressive waves of the same frequency and speed travelling in opposite directions are superposed.

Stationary wave

- Waveform does not advance.

- No net transfer of energy

- Points between two adjacent nodes are in phase.

Progressive wave

- Waveform advances with v .

- Energy is transferred in the direction of propagation.

- All points within one wavelength have different phases.

Stationary waves in fixed string (Both ends are nodes)



1st harmonic Fundamental

$$\lambda = 2L$$

$$f = \frac{v}{2L}$$

nth harmonic

(n-1)th overtone

$$\lambda = \frac{2L}{n}$$

$$f = n \left(\frac{v}{2L} \right)$$

Stationary waves in pipes with one open end



1st harmonic Fundamental

$$\lambda = 4L$$

$$f = \frac{v}{4L}$$

(2n-1)th harmonic

(n-1)th overtone

$$\lambda = \frac{4L}{2n-1}$$

$$f = (2n-1) \left(\frac{v}{4L} \right)$$

Stationary waves in pipes with two open ends



1st harmonic Fundamental

$$\lambda = 2L$$

$$f = \frac{v}{2L}$$

nth harmonic

(n-1)th overtone

$$\lambda = \frac{2L}{n}$$

$$f = n \left(\frac{v}{2L} \right)$$

* End correction: In real situations, pressure nodes and antinodes are a bit further out. Length of air column = $L + c$ (Open pipe)
OR
 $= L + 2c$ (Closed pipe)

* Measuring speed of sound:

1. Changing frequency

$$L = \frac{\lambda}{4} = \frac{v}{4f}$$

$$\Rightarrow v = 4fL$$

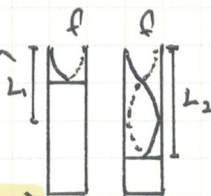


2. Changing length of air column

$$L_1 = \frac{\lambda}{4}$$

$$L_2 = \frac{3\lambda}{4}$$

$$L_2 - L_1 = \frac{\lambda}{2} \Rightarrow v = f\lambda = 2f(L_2 - L_1)$$

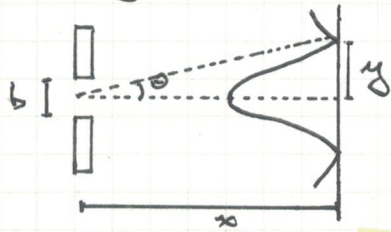


3. Using Cathode ray oscilloscope

$$v = f\lambda$$

Diffraction : Bending or spreading out of waves when they travel through a small opening or when they pass around a small obstacle.

Single slit diffraction



Slit width = b

Angle of first minima from central axis = θ

Order of minima = m

$$\sin \theta = m \frac{\lambda}{b}$$

$$\tan \theta = \frac{y}{x}$$

$$\Rightarrow \theta \approx m \frac{\lambda}{b}$$

$$\Rightarrow y = x \frac{m \lambda}{b}$$

Rayleigh criterion

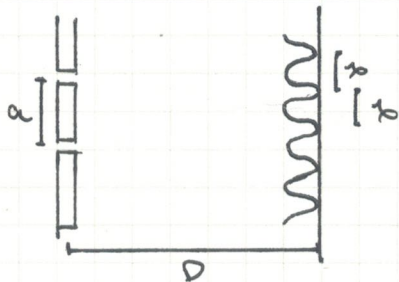
Minimum separation that two objects must have in order to be distinguished as two distinct objects = θ

$$\theta \approx \frac{\lambda}{b}$$

* The smaller the θ and the larger slit width is, the greater the resolving power.

* The smaller the wavelength of light, the higher the resolution.

Double slit interference



Slit separation = a

Distance to screen = D

Fringe separation = x

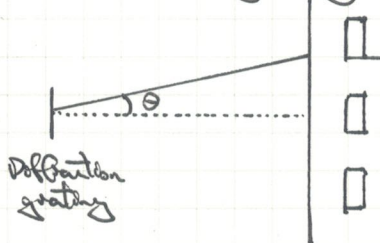
$$n \lambda = a \frac{x D}{D}$$

* $a \ll D$ for equation to be valid.

* When diffraction grating is replaced by double slit, intensity is same and less distinct peaks.

* Monochromatic light should be used.

Diffraction grating



Distance between slits = $d = \frac{\text{Length}}{\text{No. of lines}}$

Order of lines = n

Angle of line from central axis = θ

$$d \sin \theta = n \lambda$$

* Using a higher order diffracted light is more accurate as the larger angle of diffraction can be measured experimentally with a lower percentage error. However, it is dimmer.

Electric current

Electric current : Rate of flow of charge

Number density n : Number of particles per unit volume.

$$n = \frac{N}{V}$$

Drift velocity : Average velocity at which charge carriers move through a conductor when there is an electric current in the conductor.

Electric current I passing through a cross-sectional area A is defined as the rate of flow of charge Q through the area.

$$I = \frac{dQ}{dt}$$

For constant current,

$$I = \frac{Q}{t}$$

$$= \frac{Nq}{t}$$

, where N is the number of charge carriers

q is the charge of each carrier

$$= \frac{nVq}{t}$$

, where n is the number density of charge carriers,

V is the volume covered by the current

$$= \frac{nA(\Delta x)q}{(t)}$$

$$I = nAvq$$

Potential difference

: Amount of electrical energy converted to other forms of energy per unit charge.

$$V = \frac{W}{Q}$$

* Units : V or Jc^{-1}

* Work done by charges

* Earth is always at $V = 0$

* Decreases

Electromotive force

: Amount of energy transferred from non-electrical forms of energy to electrical energy per unit charge as the charge passes through a complete circuit.

$$e.m.f. = \frac{W}{Q}$$

* Current in e.m.f. source flows from low p.d. to high

* Units : V or Jc^{-1}

* Work done on charge

* Increases

Electrical power

$$P = VI = I^2R = \frac{V^2}{R}$$

Resistance : Ratio of the potential difference across the conductor to the current flowing through it.

$$R = \frac{V}{I}$$

* $R \neq \frac{dV}{dI}$, but the ratio of $V:I$

Ohm's Law : Current flowing in a conductor is directly proportional to the potential difference across it under constant physical conditions.

$$I \propto V$$

Resistivity ρ : Proportionality constant between the dimensions of a specimen of a material and its resistance such that $R = \frac{\rho L}{A}$.

* Resistance arises from collisions between electrons and metal ions.

→ ↑ Cross-sectional area ⇒ ↑ available conducting electrons ⇒ ↓ resistance

→ ↑ Length of conductor ⇒ ↑ Number of collisions ⇒ ↑ resistance

Conductors

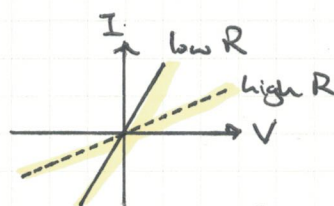
Constant temperature

• Metallic conductor

→ Is an ohmic conductor.

→ Graph is a straight through origin. Ratio of $V:I$ is constant.

→ At constant temperature, number of free electrons and rate of atomic vibration remains unchanged.

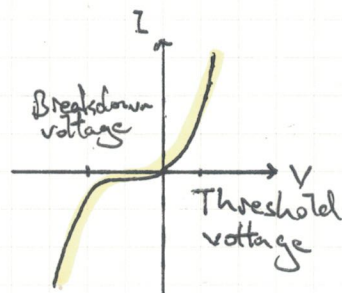


• Semiconductor diode (Forward-biased Δ)

→ As p.d. increases, resistance decreases.

→ Past threshold voltage, resistance → 0.

→ Past breakdown voltage, diode breaks down.



Varying temperature

• Filament (Metallic conductor)

→ $R \propto T$ ⇒ resistance increases with increasing temperature.

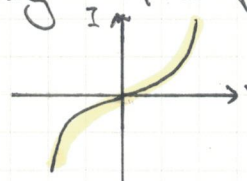
→ As I increases, power dissipated increases. Heat is generated and equilibrium temperature increases. Lattice ions vibrate more vigorously, hindering the flow of charges, increasing resistance.



• Negative temperature coefficient thermistor

→ Resistance decreases with increasing temperature.

→ As I increases, power dissipated increases. Heat is generated and equilibrium temperature increases. Mean kinetic energy of particles increases.



(i) Bonded electrons break free from bonds, increasing the number of mobile charge carriers decreasing resistance.

(ii) Lattice ions vibrate more vigorously, hindering the flow of charges. increasing resistance.

Electric Field

Coulomb's Law : The magnitude of electric force between two point charges is directly proportional to the product of their charges and inversely proportional to the square of their separation.

$$F_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-4} \text{ A}^{-2}$$

* ϵ_0 is only applicable for vacuum & air.

* Forces are action-reaction pairs.

* As $|F_E| \gg |F_g|$, when considering electric forces, F_g is negligible.

Electric field : Region of space where a charge experiences an electric force.

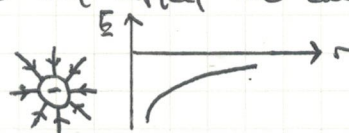
Electric field strength : Electric force per unit positive charge acting on a small positive test charge placed at the point.

$$E = \frac{F}{+q} = - \frac{dV}{dr}$$

* Test charge is small to not affect the electric field.

Point charge :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$



Parallel plate :

$$E = \frac{\Delta V}{d}$$

* Uniform acceleration

* No work is done in moving charges along equipotential lines

Electric potential energy : Work done by an external agent against the electric force in bringing the ~~point~~ ^{small} charge from infinity to the point without any change in kinetic energy.

$$U_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$* F_E = - \frac{dU_E}{dr}$$

* Opposite charges : -ve

Same charges : +ve

Electric potential : Work done per unit positive charge by an external agent in bringing the small test charge from infinity to the point without any change in kinetic energy.

$$V = \frac{U_E}{+q}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

* Scalar $\Rightarrow$ V at a point = ΣV

$$* E = - \frac{dV}{dr}$$

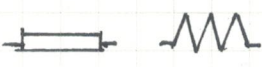
$$* \Delta V = V_{\text{final}} - V_{\text{initial}}$$

D.C. Circuits

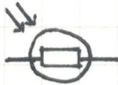
Symbols :

D.C. source :  A.C. source :  Earth : 

Resistors :

Fixed resistor : 

Variable resistor : 

Light dependent resistor (LDR) : 

* Luminance $\uparrow$
Resistance $\downarrow$

Thermistor : 

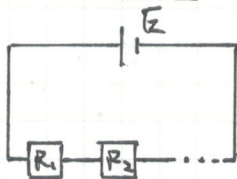
* Temperature $\uparrow$
Resistance $\downarrow$

Diode : 

Combination of resistors

* Combine circuits from the resistors furthest away from source.

Series



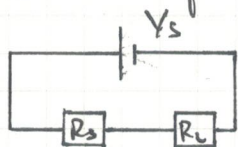
$$E = V_1 + V_2 + \dots$$

$$= IR_1 + IR_2 + \dots \quad (I \text{ is same})$$

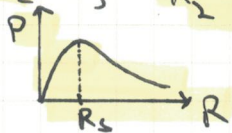
$$= IR_{\text{eff}}$$

$$\Rightarrow R_{\text{eff}} = R_1 + R_2 + \dots = \sum_{i=1}^N R_i$$

Maximum power theorem :

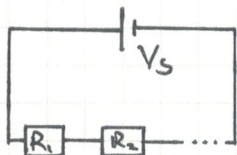


$\Rightarrow$ Maximum power is dissipated in R_L if $R_L = R_S$



Potential divider :

$\Rightarrow$ Arrangement of resistors to obtain a fraction of the potential difference applied.



$$V = IR$$

$$V_i = \frac{R_i}{R_1 + R_2 + \dots} V_S$$

$$= \frac{V_S}{R_{\text{eff}}} R_i$$

$$I_{\text{eff}} E = I_1 V + I_2 V + \dots \quad (V \text{ is same})$$

$$I_{\text{eff}} = I_1 + I_2 + \dots$$

$$\therefore I = \frac{E}{R}$$

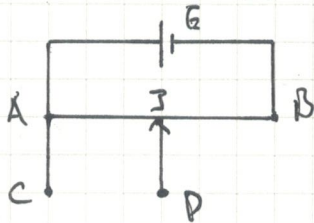
$$\Rightarrow R_{\text{eff}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots \right)^{-1}$$

* R_{eff} is smaller than resistance of the smallest resistor.

Potentiometer

$\Rightarrow$ Uses principle of potential divide to measure potential difference, e.m.f. of unknown cell, internal resistance of cell.

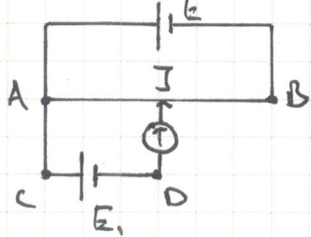
Potential difference :



$$\frac{V_{AJ}}{V_{AB}} = \frac{R_{AJ}}{R_{AB}} = \frac{L_{AJ}}{L_{AB}}$$

$$\Rightarrow V \propto L$$

e.m.f. of unknown cell :

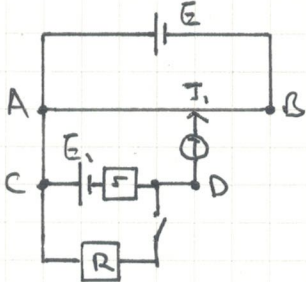


$$\Rightarrow \frac{V_{AJ}}{R_{AJ}} = \frac{V_{AB}}{R_{AB}}$$

$$\Rightarrow \frac{E_1}{E} = \frac{L_{AJ}}{L_{AB}} = \frac{R_{AJ}}{R_{AB}}$$

p.d. across CD
=
p.d. across AJ
Hence $I=0$

Internal resistance of cell :



Switch open : $\frac{E_1}{E} = \frac{L_{AJ}}{L_{AB}} \Rightarrow L_{AJ} = \frac{E_1}{E} L_{AB}$

Switch closed :

Current in standalone circuit $I = \frac{E_1}{R+r}$

p.d. across $R = E_1 - Ir$

$$= \frac{R}{R+r} E_1$$

p.d. across $AJ_2 = \frac{L_{AJ}}{L_{AB}} E$

* Potential meter at balance point does not draw current from the primary circuit whereas a practical voltmeter does.

Combination of battery cells

Series

$$V_{eff} = V_1 + V_2 + \dots = \sum_{i=1}^N R_i$$

Capacity is constant

Parallel

$$V_{eff} = V_1 = V_2 = V_i$$

Capacity is multiplied

Electromagnetism

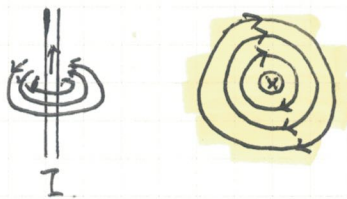
Magnetic field : Region of space where a magnetic pole, moving charged particle or current-carrying conductor will experience a magnetic force.

* Only moving charges generate magnetic fields and experience magnetic force.

Magnetic flux density B : Force per unit current per unit length acting on an infinitely long current-carrying conductor placed perpendicular to the magnetic field.

The tesla : Magnetic flux density required for a long straight conductor carrying 1A of current in an uniform magnetic field normal to the wire to experience 1 of force per metre.

Long straight current-carrying conductor



$$B = \frac{\mu_0 I}{2\pi d}$$

d : Perpendicular distance between point and wire

Flat circular coil with turns



$$B = \frac{\mu_0 N I}{2r}$$

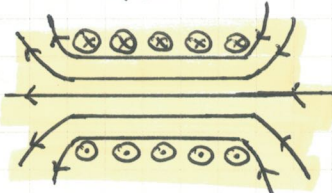
r : Radius of coil

N : Number of turns

B : Magnetic flux density at centre of the coil

Long straight solenoid

* $L \gg r$



$$B = \mu_0 n I$$

n : Number of turns per unit length = $\frac{N}{L}$

B : Magnetic flux density at the inside of the solenoid

* B can be increased by placing a soft iron core within the solenoid.

* Ferromagnetic cores can be magnetised easily. The alignment of magnetic domains strengthens the overall magnetic field, increasing magnetic flux density.

Magnetic forces

On moving charge

$$F_B = qvB \sin \theta$$

q : charge

v : Speed of moving charge

θ : Angle between B and v

* If velocity of charged particle is perpendicular to the magnetic field, charged particle undergoes circular motion.

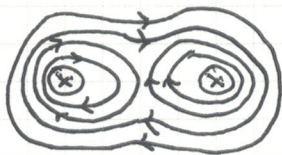
On straight current-carrying wire

$$F_B = BIl \sin \theta$$

l : Length of conductor

θ : Angle between I and B

Between two long parallel current carrying conductors



Same direction attract

$$\therefore B = \frac{\mu_0 I}{2\pi d}$$

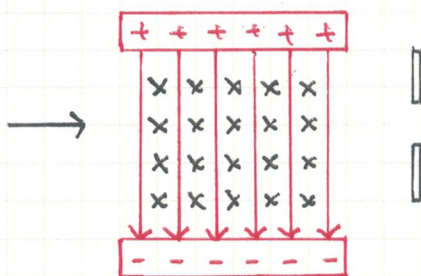
$$\therefore F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$$



Opposite direction repel

Velocity selector

→ Electric and magnetic fields are set up such that F_E and F_B are in opposite directions.



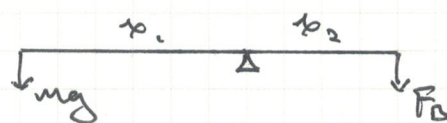
→ Only particles with $F_{net} = 0 \Rightarrow qvB = qE$

Hence, all particles that pass through the slit must have the same velocity

Current balance

→ Used to determine magnetic flux density

$$mgx_1 = BIlx_2$$



* Shapes : Helix :  Spiral : 

Electromagnetic Induction

Magnetic flux Φ : Product of magnetic flux density B and the area normal to the field through which the field is passing.

$$\Phi = BA \cos \theta$$

1 Weber : Magnitude of magnetic flux through an area of 1 m^2 when a magnetic field of 1 T acts perpendicularly into the area.

Magnetic flux linkage : Product of number of turns of the coil and the magnetic flux Φ passing through the coil.

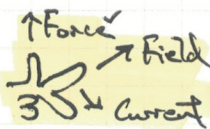
$$N\Phi = NBA \cos \theta$$

Faraday's law of electromagnetic induction : e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage.

$$E = - \frac{d(N\Phi)}{dt} = - \frac{d(NBA \cos \theta)}{dt}$$

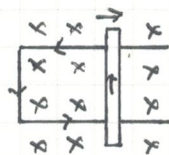
* Negative sign is due to Lenz's law

Lenz's Law : Direction of induced e.m.f. is such that it produces an effect that opposes the change causing it.

Fleming's left-hand rule : 

right-hand rule : 

Motional e.m.f.



$$|E| = \frac{d\Phi}{dt} = \frac{\Delta\Phi}{\Delta t} = Bl \frac{\Delta x}{\Delta t} = Blv$$

$$\Rightarrow E = IR = Blv$$

$$I = \frac{Blv}{R}$$

Applications :

- AC generator
- Transformers
- Electric guitar



Eddy currents

- ↳ Loops of electrical current induced by the changing magnetic field.
- ↳ According to Lenz's Law, direction of eddy currents is such that it produces a magnetic field to oppose the change in magnetic field. This gives rise to a repulsive force.

Alternating Currents

Alternating current : Current that varies in magnitude and direction periodically.

Sinusoidal : $I = I_0 \sin \omega t$ OR $I = I_0 \cos \omega t$

* Mean I and $V = 0$

Root-mean-square : The root-mean-square value of an alternating current / voltage is the value of a steady direct current that would produce the same average power in a given resistor.

$$P_{d.c.} = \langle P_{a.c.} \rangle$$

$$\langle I_{d.c.}^2 \rangle R = \langle I_{a.c.}^2 \rangle R$$

$$I_{d.c.} = \sqrt{\langle I_{a.c.}^2 \rangle} = I_{rms}$$

To find r.m.s. value :

Square $\rightarrow$ Mean $\rightarrow$ Root

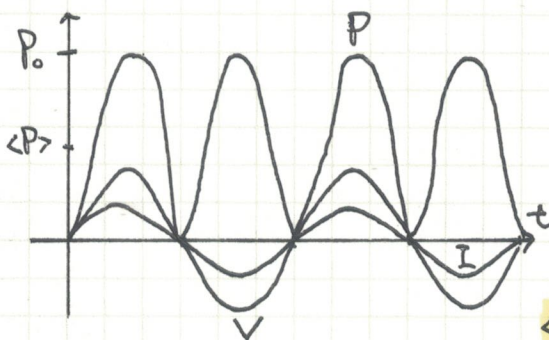
1. Square I or V

2. Find mean by $\frac{\text{Area within 1 period}}{T}$

3. Root. $\sqrt{\frac{\int_a^b f(t) dt}{b-a}}$

$$\langle P \rangle = I_{rms}^2 R = \frac{V_{rms}^2}{R} = I_{rms} V_{rms}$$

For sinusoidal a.c. : $I_{rms} = \frac{I_0}{\sqrt{2}}$



$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

$$\langle P \rangle = \frac{1}{2} P_0$$

* Power-time graph is sinusoidal.

Transformer : $\exists \parallel \exists$

Transformer cannot increase power!

$$P_p = P_s$$

$$V_p I_p = V_s I_s$$

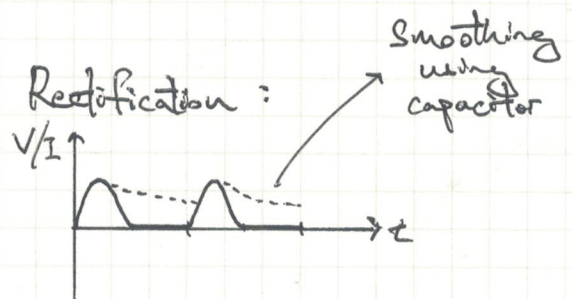
$$\frac{V_s}{V_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$$

* For ideal transformer

Energy losses :

- Eddy currents
- Windage resistance
- Hysteresis losses

Rectification :



Forward biased $\rightarrow$ Zero resistance

Reversed biased $\rightarrow$ Infinite resistance

Quantum Physics

Particulate nature of light

Photon : A quantum of electromagnetic energy.

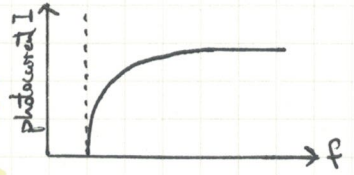
Planck's equation : $E = hf = h \frac{c}{\lambda}$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

* **Total energy of monochromatic light** = Nhf

Power of monochromatic light = $\frac{E}{t} = \frac{Nhf}{t}$

Intensity of monochromatic light = $\frac{P}{A} = \frac{Nhf}{tA} \Rightarrow I \propto \left(\frac{N}{t}\right)(hf)$



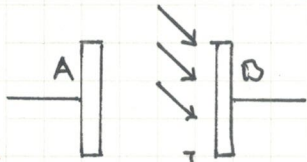
To maintain intensity if energy increases, $\frac{N}{t}$ must decrease.

Photoelectric effect : Emission of electrons from a metal surface

when the surface is irradiated with

electromagnetic radiation of a high enough frequency.

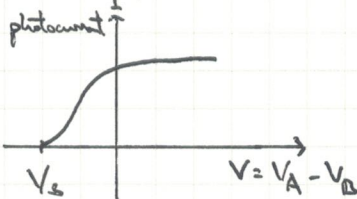
$$hf = \phi + \frac{1}{2}mv_{\max}^2$$



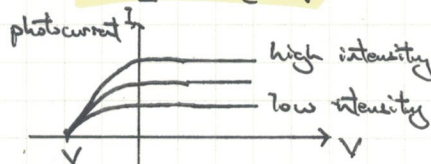
$V_A > V_B$: Electrons are attracted to A, producing a steady current

$V_D > V_A$: Electrons experience a repulsive force and decelerate.

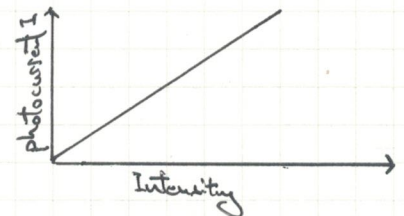
$$\Delta KE = e \Delta V$$



$$V_s = \frac{hf}{e} - \frac{\phi}{e}$$



$\Rightarrow V_s$ is independent of intensity.



With increasing number of photons per unit area and time, more electrons can be liberated.

Experimental observations

- **Instantaneous.**

- Electrons are only emitted when $f \geq f_0$.

Emission is independent of intensity / amplitude.

- Max KE of emitted electron is dependent on f and independent of intensity.

- Photocurrent $\propto$ Intensity.

Classical theory

✗ Electrons absorb energy over a period of time. Time taken $\propto$ Intensity

✗ Electrons are only emitted when they have enough intensity / amplitude.

✗ KE of electron should depend on energy of light that is proportional to intensity and amplitude.

✓ High intensity of light forces more electrons

Quantum theory

- Energy transfer is one-to-one. Energy of a photon cannot be shared.

- **Work function** : Minimum energy required to eject an electron from the surface of a metal.

$$\phi = hf_0$$

- $KE_{\max} \propto f$
 $KE_{\max} = hf - \phi$

$$I = \left(\frac{N}{t}\right) \left(\frac{hf}{A}\right)$$

Wave-particle duality

momentum of light $p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$

de Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$

* Diffraction $\propto \frac{\lambda}{b} \Rightarrow \propto \frac{h}{mvb}$

Heisenberg Uncertainty Principle : $\Delta p \Delta x \geq h \left| \frac{\Delta x}{\Delta p} \right|$

Energy levels in atoms

Ground state : Lowest state of an atom where all electrons in an atom occupy the lowest possible energy states.

Excited state : State of an atom where one or more of its electrons are not occupying the lowest possible energy states.

Electron transition : electron excite - absorb photon

* Given f/λ of highest and lowest state,

$$hf = \Delta E = |E_f - E_i|$$

electron deexcite - emit photon

$$f_2 = f_1 - f_0$$

$$\lambda_2 = \frac{\lambda_1 \lambda_0}{\lambda_2 - \lambda_1}$$

Emission and absorption line spectra

Emission line spectrum : Series of distinct colored lines against a dark background.

$$\text{number} = nC_2$$

Absorption line spectrum : Series of distinct dark lines against a continuous spectrum.

$$\text{number} = n - 1$$

* Transition to excited state is mainly from ground state.

X-ray spectra : Produced when electrons collide with a metal target.

* A conducting material is needed to produce a large p.d. to accelerate the electrons, hence a metal target.

Cut-off wavelength : - Electron completely stops and entire KE is lost as a single photon.

$$hf_{\text{max}} = h \frac{c}{\lambda_{\text{min}}} = eV$$

* With higher tube voltage, max KE of electrons is higher, and as it completely stops, a photon with a higher energy is emitted.

Characteristic lines M, L and K : - Electrons collide with electron in inner shell

* Dependent on the energy transitions. creating a vacant spot.

* Independent of voltage.



- Electron from a higher shell transit to the vacancy, emitting a photon.

Continuous Bremsstrahlung

- Electrons are accelerated, decelerated and deflected, losing KE in form of photons.

- Energy lost in terms of photons can be any amount less than the max KE.

* In this process, more electrons with higher KE

